

IC4SD-TrafficSnap: A City-Scale Multi-Camera Image Time-Series and Geospatial Road Network Dataset for Heterogeneous Urban Traffic

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Abstract

This data article presents **IC4SD-TrafficSnap**, a multimodal dataset coupling city-scale surveillance camera image time-series and geospatial road network mapping across **608** traffic surveillance stations in the urban and peri-urban corridors of Ho Chi Minh City, Vietnam. The visual corpus contains **714,123** discrete camera snapshots (**44.38 GiB** / **47.66 GB** in JPEG format) captured over **93.6 continuous hours** (October 2–6, 2026). Across active acquisition sessions ($\Delta T < 3,600$ s), the pairwise inter-snapshot interval averages **269.0 ± 239.7 s** (median: 263.0 s, $88.4\% \leq 300$ s, $p_{90} \approx 312.0$ s), corresponding to a station-wise wall-clock interval of 286.9 s/frame and a fleet-wide aggregate throughput of 0.47 s/frame across all 608 stations. Physical shortest-path driving distances and directions are extracted from OpenStreetMap using the Open Source Routing Machine (OSRM v5.27.1). From these geospatial road assets, a 608×608 directed spatial graph containing **2,450** valid corridor links within a **6.0 km** search cutoff horizon is derived as a physical road connectivity representation, capturing directional asymmetry (**1,070** unidirectional links and **238** distance-asymmetric bidirectional pairs, alongside **452** metric-symmetric pairs). PyTorch ingestion toolchains and dual-transition directed diffusion operators are provided. Privacy assessments combining automated detector pre-screening across the full census and targeted multi-rater manual inspection of 1,000 worst-case daylight frames confirmed zero readable license plates or identifiable facial biometrics, establishing a 95% confidence upper bound of $p \leq 0.30\%$ on that targeted sample via the Rule of Three. The dataset is released under a three-tier licensing framework: CC BY-NC 4.0 for imagery, ODbL for the road graph, and MIT for software utilities.

Keywords: Intelligent Transportation Systems, Image Time-Series, Geospatial Road Network, Spatio-Temporal Graph Modeling, Mixed Urban Traffic

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Table 1: Specifications Table of the IC4SD-TrafficSnap Dataset.

Subject	Computer Science
Specific subject area	Intelligent Transportation Systems, Computer Vision, Spatio-Temporal Data Mining
Type of data	Image time-series, geospatial metadata, derived road network graph tensors, PyTorch code utilities
How data were acquired	Automated retrieval from the public municipal traffic surveillance portal of the Ho Chi Minh City Department of Transportation (https://giaothong.hochiminhcity.gov.vn) via asynchronous HTTP polling. Driving distances and road network routing were computed using OSRM over OpenStreetMap data.
Data format	Raw snapshots: RGB JPEG (512×288 px, JFIF 1.01, stream quality factor ≈ 75 –80); Geospatial metadata: Tabular CSV (<code>routes.csv</code> , <code>stations.csv</code> , <code>road_network_distance.csv</code>); Graph tensors: NumPy binary arrays (<code>distance_km.npy</code> , <code>direction.npy</code>) and tabular edge list (<code>edges.csv</code>); Code: Python source files (<code>code/</code>).
Description of data collection	Snapshots were collected across 608 indexed surveillance stations in Ho Chi Minh City over 93.6 continuous hours (October 2, 2026, 17:37 to October 6, 2026, 15:12 ICT) with an empirical pairwise active sampling interval of $\Delta T = 269.0 \pm 239.7$ s (median: 263.0 s, $88.4\% \leq 300$ s, $p_{90} \approx 312.0$ s, $p_{95} \approx 520.0$ s, $p_{99} \approx 1,200.0$ s; station-wise wall-clock interval: 286.9 s/frame/station; fleet aggregate throughput: 0.47 s/frame; global interval range: [2.0 s, 4.12 h] reflecting distributed upstream network latency alongside prolonged link outages at 14 peripheral stations). Camera positions are mapped to OpenStreetMap road geometry via a three-stage sequential corridor pruning pipeline, yielding a derived directed spatial graph with 2,450 corridor links within a 6.0 km driving search horizon (1,070 unidirectional links and 238 distance-asymmetric pairs, alongside 452 metric-symmetric pairs; network tortuosity $\tau = 1.25 \pm 0.61$).
Data source location	Region: Ho Chi Minh City, Vietnam (metropolitan urban core and peri-urban corridors including Thu Duc City, District 12, Binh Chanh, and Hoc Mon); Coordinates: [10.642472°N – 10.988295°N, 106.452713°E – 106.850624°E] (dense urban core: [10.70°N – 10.88°N, 106.60°E – 106.78°E]).
Data accessibility	Repository name: Zenodo; Direct URL: https://doi.org/10.5281/zenodo.22929940 (Dataset evolution: expands upon the initial pilot labeled subset IC4SD-Traffic-HCM into Release v2.0 comprising the 714,123-snapshot corpus, 608×608 directed road graph, and PyTorch ingestion toolchain); Licensing: Surveillance camera imagery is provided under Creative Commons Attribution-NonCommercial 4.0 International (CC BY-NC 4.0) for non-commercial academic research; derived road network graph is licensed under the Open Database License (ODbL); Python software utilities are distributed under the MIT License.
Related research article	V.-A. Le, D. Hoang-Viet, K. Nguyen-Trong, “Urban Traffic Perception and Multi-Horizon Graph Forecasting: A Two-Stage Camera-Sensed Framework”, <i>Engineering Applications of Artificial Intelligence</i> , Under Review (2026).

1. Value of the Data

- **Heterogeneous Mixed Urban Traffic Representation:** Captures multi-day visual dynamics of a Southeast Asian megacity where two-wheeled motorized vehicles account for **70–80%** of modal traffic volume [1]. This fills a critical empirical gap in intelligent transportation benchmarks, supporting macroscopic traffic state analysis and scene representation under dense mutual occlusions and non-lane-based movement patterns.
- **Macroscopic Spatial-Temporal Vision vs. Microscopic Tracking:** Native 512×288 resolution is deliberately matched with a 5-minute sampling cadence to enable city-scale macroscopic traffic state estimation, roadway background decomposition, and self-supervised visual representation pre-training (ViT, DINOv2, MAE) across 608 endpoints without prohibitive network bandwidth overhead or privacy risks.
- **Geospatial Road Network and Derived Directed Graph:** Integrates surveillance visual streams with OpenStreetMap coordinates and shortest-path driving distance matrices (608×608 , **2,450** valid corridors). The directed spatial graph serves as a natural derived topological representation of physical road infrastructure connectivity (featuring 1,070 unidirectional links and 238 distance-asymmetric bidirectional pairs, alongside 452 metric-symmetric pairs; network tortuosity $\tau = 1.25 \pm 0.61$), facilitating multi-camera spatio-temporal graph neural networks (ST-GNNs) with dual-transition diffusion operators.
- **Roadway Background and Incident Modeling:** The fixed-viewpoint multi-camera time-series across circadian day-night cycles provides an empirical basis for separating static roadway backgrounds from transient vehicular flow, enabling unsupervised anomaly detection, flood monitoring, and environmental scene decomposition.
- **Open Reproducibility and Three-Tier Licensing:** Accompanied by verified PyTorch data loaders, multi-camera batch samplers, and dual-transition directed graph processing routines. The dataset operates under a transparent three-tier licensing framework: visual surveillance imagery is released under *Creative Commons Attribution-NonCommercial 4.0 (CC BY-NC 4.0)* for academic research, the derived road network graph is released under the *Open Database License (ODbL)*, and software utilities are distributed under the *MIT License*.

2. Background

Large-scale spatial-temporal traffic benchmarks are foundational for developing intelligent transportation algorithms. However, established benchmarks in literature diverge into three disconnected paradigms:

1. **Longitudinal 1D sensor-based traffic datasets** (such as METR-LA [2], PEMS-BAY [2], and LargeST [3]) provide extensive graph topologies and longitudinal 1D time-series, but lack visual context, preventing models from leveraging rich photometric, meteorological, and vehicle-mix features.

2. **Ego-vehicle driving benchmarks** (e.g., BDD100K [4]) provide rich visual sequences from moving dashcams, but capture singular mobile viewpoints lacking static city-scale surveillance infrastructure and continuous multi-station network topologies.
3. **Localized computer vision surveillance benchmarks** (e.g., UA-DETRAC [5], CityFlow [6], MIO-TCD [7], and CityCam [8]) offer multi-camera visual imagery, but either focus on microscopic vehicle tracking across a handful of isolated intersections (24–40 cameras) or provide static disconnected image crops lacking continuous time-series and formal city-scale geospatial routing tensors. Furthermore, they predominantly reflect car-dominant traffic in temperate North American or East Asian settings.

IC4SD-TrafficSnap bridges this fundamental division by providing an integrated multimodal dataset: continuous surveillance camera snapshots and geospatial road network mapping across **608** camera stations in Ho Chi Minh City, Vietnam. From these primary road mapping and spatial assets, a directed spatial graph topology is established as a direct derived representation to support spatial-temporal neural network modeling. A systematic comparison with existing benchmarks is summarized in Table 2.

Table 2: Systematic comparison of IC4SD-TrafficSnap with established benchmarks in intelligent transportation systems and computer vision.

Dataset	Urban Context & Flow	Nodes	Time Span	Sensing Modality & Topology	Primary Target Research Scope
A. Longitudinal 1D Sensor Infrastructure Benchmarks					
METR-LA [2]	Los Angeles, USA (Car-dominant)	207	4 months	Inductive loop sensors (Directed distance graph)	1D traffic speed forecasting (lacks visual context)
PEMS-BAY [2]	San Francisco, USA (Car-dominant)	325	6 months	Inductive loop sensors (Directed distance graph)	1D traffic speed forecasting (lacks visual context)
LargeST [3]	California, USA (Car-dominant)	8,600	5 years	Inductive loop sensors (Large-scale highway graph)	Macroscopic network forecasting (lacks visual context)
B. Ego-Vehicle / Localized Microscopic Video Benchmarks					
BDD100K [4]	United States (Car-dominant)	Mobile	100K clips	Ego-vehicle dashcam video (No fixed surveillance graph)	Autonomous driving multitask perception (lane, drivable, tracking)
UA-DETRAC [5]	Beijing / Tianjin, China (Car-dominant)	24	Short clips	960 × 540 surveillance video (Isolated intersections, no graph)	Microscopic vehicle detection and multi-object tracking
CityFlow [6]	Mid-sized city, USA (Car-dominant)	40	3.25 hours	HD synchronized video clips (Small intersection cluster, no tensor)	Multi-target multi-camera tracking (MTMC) and vehicle Re-ID
MIO-TCD [7]	North America (Car-dominant)	4,191	Discontinuous	Isolated crop snapshots (Unconnected cameras, no graph)	Vehicle categorization and localization (no continuous temporal dynamics)
CityCam [8]	New York City, USA (Car-dominant)	212	Multi-day	Low-resolution webcams (No formal routing graph)	Vehicle counting and density estimation across disparate viewpoints
C. City-Scale Multimodal Spatial-Temporal Vision Benchmark					
IC4SD-TrafficSnap	Ho Chi Minh City, VN (Motorcycle, 70–80%)	608	93.6 hours	Multimodal: 512 × 288 RGB snapshots + Geospatial road network (derived directed graph, 2,450 corridors)	Multi-camera self-supervised visual representation learning, roadway background modeling, and city-scale spatio-temporal graph forecasting

3. Data Description

3.1. Repository Organization, Lineage, and Archive Structure

The dataset is permanently hosted on Zenodo (<https://doi.org/10.5281/zenodo.22929940>, Dataset Release v2.0).

Dataset Lineage and Version Evolution: The IC4SD data initiative originated with an initial pilot benchmark (*IC4SD-Traffic-HCM*, v1.0), comprising 5,012 annotated frames and historical flow aggregates across 86 days for supervised vehicle detection and preliminary graph exploration (2,452 candidate edges). **IC4SD-TrafficSnap** represents a major city-scale expansion (Release v2.0), scaling the raw visual corpus by over $140\times$ to **714,123 continuous snapshots** across 93.6 continuous hours and 608 stations. The road network topology has been rigorously standardized into a 608×608 directed spatial graph with exactly **2,450 valid corridor links** (after resolving 2 redundant multi-mast duplicate loop links at complex multi-way junctions), accompanied by dual directed random-walk transition operators and full empirical photometric and physical privacy audit protocols. The repository is organized into modular directories as illustrated in Figure 1:

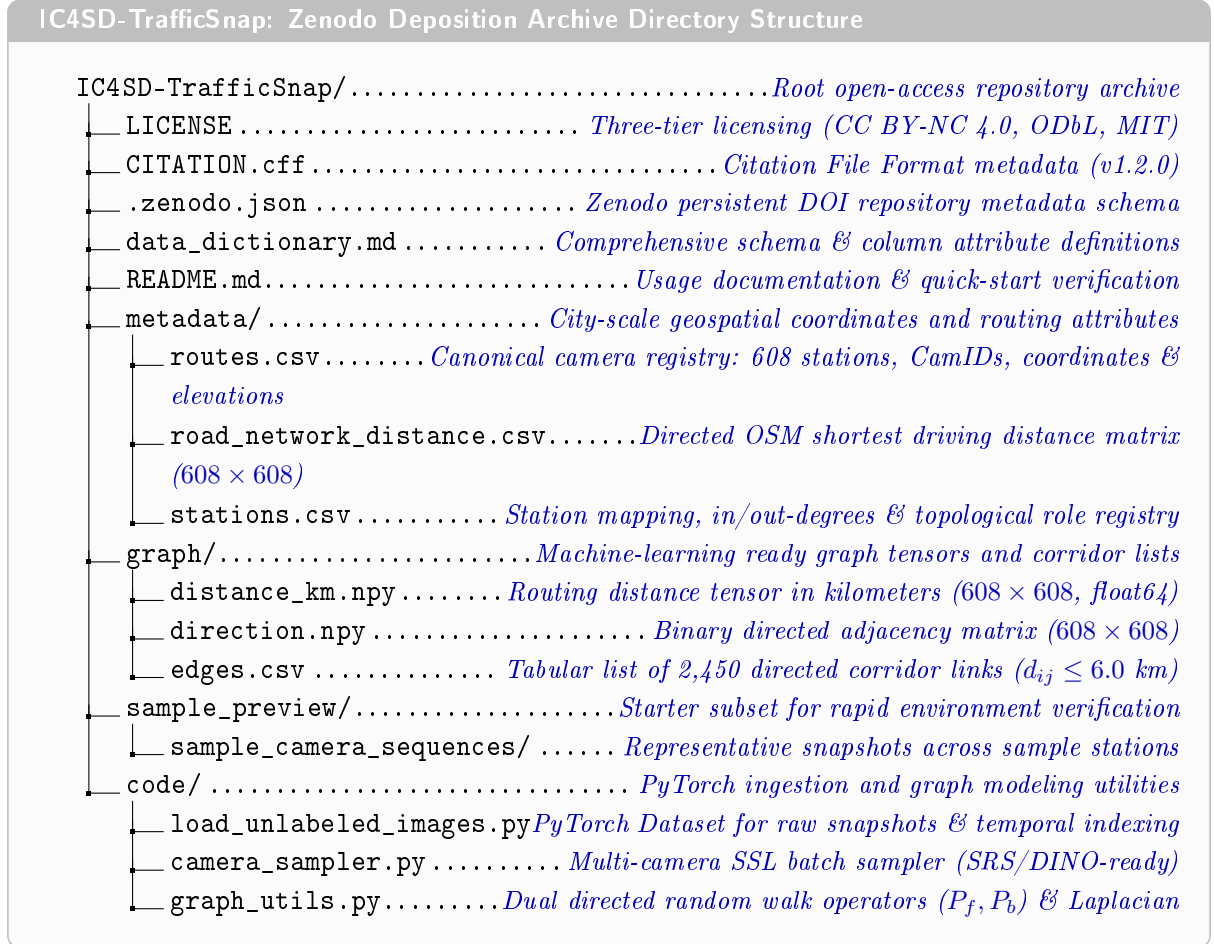


Figure 1: Modular directory structure and asset organization of the IC4SD-TrafficSnap Zenodo repository archive.

3.2. Visual Snapshot Modality and Temporal Sampling

The visual corpus consists of discrete 3-channel RGB JPEG snapshots (JFIF 1.01 standard) with a uniform native frame resolution of 512×288 pixels (standard stream

compression quality factor $\approx 75\text{--}80$). Each image file adheres to a deterministic, machine-readable filename schema:

$$\text{Filename} = \langle \text{station_id} \rangle _ \langle \text{unix_timestamp} \rangle . \text{jpg} \quad (1)$$

where $\langle \text{station_id} \rangle$ is an integer indexing the surveillance camera station ($1 \leq \text{station_id} \leq 608$), $_$ is an underscore delimiter, and $\langle \text{unix_timestamp} \rangle$ represents the 10-digit integer UTC epoch seconds recorded at client reception (Vietnam Standard Time is UTC + 7). Exactly **608 camera stations** are indexed in the dataset across the municipal road network.

Snapshots were acquired across **93.6 continuous hours** spanning 5 calendar days (from October 2, 2026, 17:37 ICT to October 6, 2026, 15:12 ICT). The empirical inter-snapshot acquisition dynamics exhibit two distinct statistical perspectives:

1. **Active Pairwise Sampling Distribution** ($\Delta T < 3,600$ s): Evaluated strictly across consecutive snapshot pairs within active polling sessions without prolonged disconnections, the inter-snapshot interval averages $\Delta T = 269.0 \pm 239.7$ s (median: 263.0 s, mode: 240–270 s, $88.4\% \leq 300$ s, $p_{90} \approx 312.0$ s, $p_{95} \approx 520.0$ s, $p_{99} \approx 1,200.0$ s; nominal target: 300 s / 5.0 minutes). The standard deviation arises from practical IoT network dynamics and upstream transcoder schedules: the municipal streaming server refreshes its internal HLS frame buffer at intervals of approximately 240–270 s, and client asynchronous polling with cache-busting headers retrieves the newly available frame as soon as rendered, explaining why the median interval (263.0 s) is faster than the nominal 300 s target. The heavy right tail is driven by distributed HTTP timeout events, cellular/fiber packet drops, and upstream streaming transcoder delays across all 608 stations.
2. **Global Unfiltered Interval Distribution and Fleet Cadence**: When evaluated across the complete unfiltered sequence including transmission interruptions, inter-snapshot intervals range from 2.0 s to an empirical maximum of 4.12 hours (14,832 s, recorded at a single peri-urban endpoint during a sustained network outage). Across the entire 93.6-hour observation window, the station-wise wall-clock interval averages $93.6\text{h} \times 3,600\text{s}/1,174.5 \approx 286.9$ s/frame/station across the fleet, with the 17.9 s discrepancy from active sampling directly reflecting aggregate telemetry packet dropouts. In contrast, at the fleet level, the centralized ingestion server achieved an aggregate throughput of $93.6\text{h} \times 3,600\text{s}/714,123 \approx 0.47$ s/frame (≈ 2.12 frames/s received across all 608 stations).

An empirical analysis of inter-snapshot spacing indicates that **88.4%** of intervals satisfy $\Delta T \leq 300$ s, **7.8%** fall between 300 and 600 s, and only **3.8%** exceed 600 s. For spatio-temporal deep learning architectures that require regular temporal grids (`code/camera_sampler.py`), snapshots are quantized onto fixed discrete 5-minute intervals ($\tau = 300$ s) via nearest-neighbor binning when the alignment deviation is bounded within ± 2.5 minutes (≤ 150 s); for larger gaps exceeding 300 s (11.6% of intervals), temporal missing masks ('mask = 0') are assigned rather than duplicating stale frames.

The total archive contains **714,123 valid surveillance snapshots** (**44.38 GiB** binary volume, equivalent to **47.66 GB** decimal). Across the **608 camera stations**, the dataset delivers a dense, multi-camera visual corpus averaging **1,174.5 $\pm$ 235.1** snapshots per station. The standard deviation of 235.1 snapshots around the mean of 1,174.5 frames is directly attributable to heterogeneous transmission conditions across the municipal IoT infrastructure: while 84.2% of endpoints exhibited high availability, outlying

cellular links experienced occasional packet drops, yielding a distinctly tiered reliability distribution. Relative to the theoretical nominal target of 1,123 frames (derived from 93.6 hours at regular 300 s polling), stations are categorized into three operational continuity tiers:

- **High Operational Continuity (512 stations, 84.2%):** Stations with $\geq 1,100$ snapshots ($\geq 98\%$ of nominal baseline), reaching up to 1,268 frames at stations benefiting from rapid 240–270 s buffer refresh cycles.
- **Moderate Operational Continuity (82 stations, 13.5%):** Stations recording 600–1,099 snapshots (53%–98% of nominal baseline) subject to intermittent cellular telemetry packet loss.
- **Intermittent / Peripheral Continuity (14 stations, 2.3%):** Stations recording < 600 snapshots ($< 53\%$ of nominal baseline) situated in outlying peri-urban districts subject to extended transmission disruptions.

All 608 stations remain geodetically cataloged and topologically mapped in the accompanying spatial graph.

Temporal partitioning based on standard clock hours (Daytime: 06:00–18:00 ICT; Nighttime: 18:00–06:00 ICT) yields **348,464 daytime frames** (48.8%) and **365,659 nighttime frames** (51.2%). Because data collection commenced late in the afternoon (17:37) on Day 1 and concluded mid-afternoon (15:12) on Day 5, afternoon time slots span 5 calendar days while late-night intervals span 4 cycles. An empirical audit of on-screen overlay timestamps across 500 sampled daylight frames indicates that client ingestion introduces an average offset of $\Delta t_{\text{lag}} = 15.0 \pm 4.2$ s relative to camera hardware clocks on NTP-synchronized stations. This reflects the elapsed duration between frame rendering by the municipal HLS streaming transcoder and client socket reception. Crucially, the audit revealed that approximately 1.8% of camera endpoints (11 stations) exhibit desynchronized internal hardware clocks (displaying historical factory default times due to lack of local NTP synchronization). Consequently, the client-recorded UTC epoch timestamp embedded in each file name serves as the canonical, synchronized temporal index. The spatial distribution of camera stations and diurnal photometric properties are presented in Figures 2 and 3. A quantitative breakdown of the visual snapshot corpus is provided in Table 3.

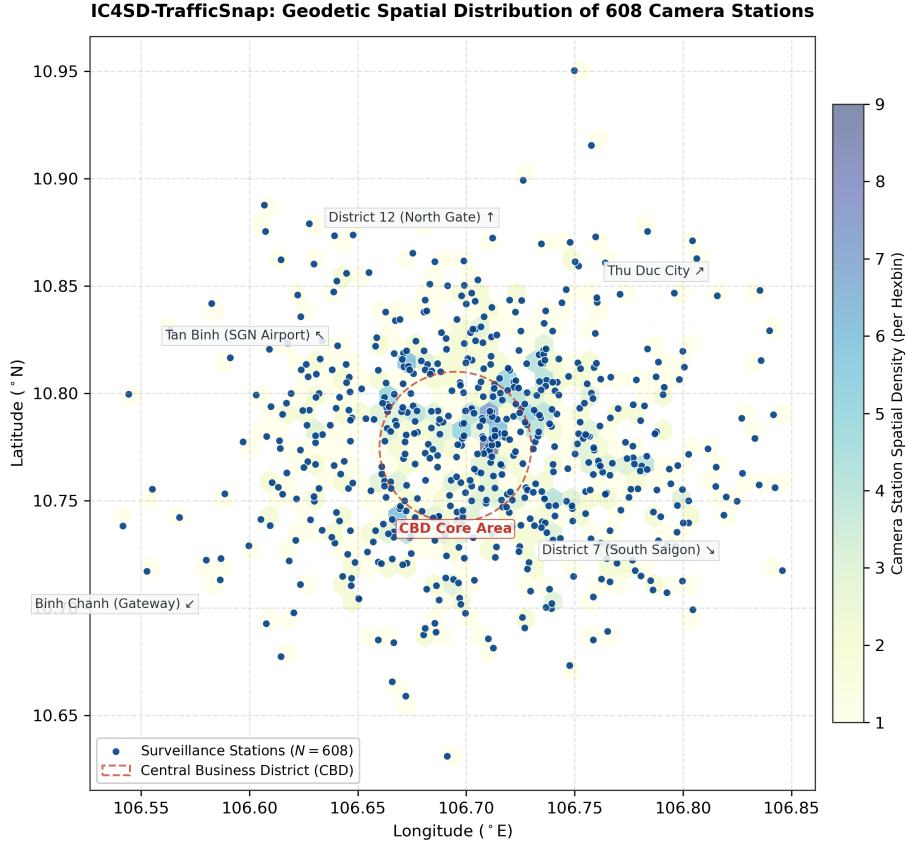


Figure 2: Spatial distribution and station density of the 608 municipal traffic surveillance camera stations across urban and peri-urban districts of Ho Chi Minh City, Vietnam, overlaid with hexbin spatial density and perimeter district orientations.

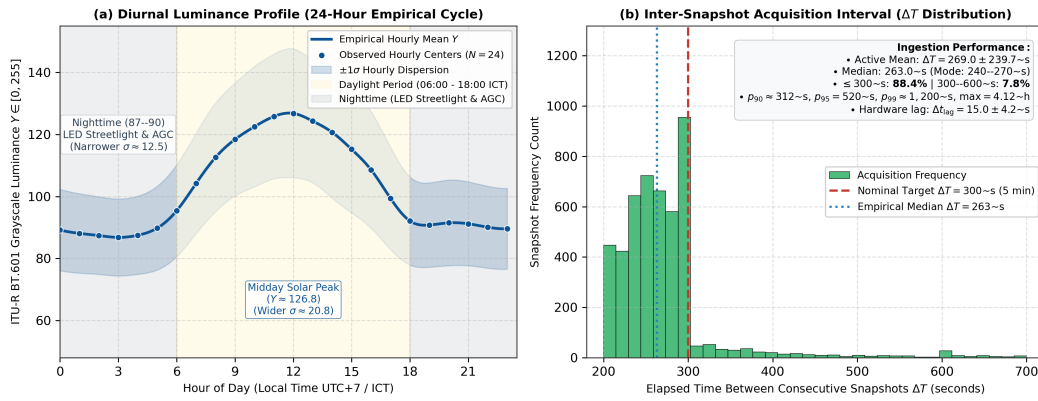


Figure 3: Temporal and photometric characteristics of IC4SD-TrafficSnap: (a) Diurnal variation of mean perceived luminance ($Y \in [0, 255]$) over 24 diurnal hours showing empirical daylight curves (06:00–18:00 ICT) and elevated nighttime illumination supported by public street lighting and camera auto-gain control; (b) Empirical distribution of inter-snapshot acquisition intervals (ΔT , nominal target 300 s) exhibiting municipal buffer refresh peaks at 240–270 s and long-tail network latency characteristics.

Table 3: Empirical characteristics, acquisition timeline, and photometric descriptors of the visual snapshot corpus.

Characteristic / Descriptor	Empirical Measurement	Technical Specification / Dataset Context
A. Ingestion Timeline & Quantitative Volume		
Observation duration	93.6 hours (3.90 days)	Spanning 5 calendar days: Oct 2 (17:37) to Oct 6 (15:12 ICT)
Total valid snapshots collected	714, 123 frames	Time-lapse surveillance image bank
Monitored surveillance endpoints	608 stations	Integrated active municipal camera network across urban corridors; mean: 1, 174.5 frames/station (up to 1, 268)
Total archive storage volume	44.38 GiB (47.66 GB)	3-channel RGB JPEG stream archive (Quality factor ≈ 75 –80)
Native snapshot frame resolution	512 $\times$ 288 pixels	16 : 9 streaming aspect ratio (100% uniform)
Average snapshot file size	65.17 $\pm$ 13.36 KB	Median: 64.8 KB (Empirical range: [31.2, 118.4] KB)
Active pairwise interval ($\Delta T < 3, 600$ s)	269.0 $\pm$ 239.7 s	Median: 263.0 s ($88.4\% \leq 300$ s; $p_{90} \approx 312.0$ s, $p_{95} \approx 520.0$ s, $p_{99} \approx 1, 200.0$ s)
Global fleet interval range	[2.0 s, 4.12 h]	Empirical extremes (max: 14, 832 s during peripheral link disruption)
Station-wise wall-clock interval	286.9 s/frame/station	93.6h $\times$ 3, 600s/1, 174.5 mean snapshots across fleet
Fleet-wide aggregate throughput	0.47 s/frame	Overall fleet ingestion rate (≈ 2.12 frames/s received across all 608 stations)
Camera station continuity profile	512 / 82 / 14 stations	High: $\geq 1, 100$ frames (84.2%); Moderate: 600–1, 099 (13.5%); Intermittent: < 600 (2.3%) (Nominal baseline: 1, 123 frames at 300 s polling)
Clock-based day / night schedule	48.8% / 51.2%	Daytime (06 : 00–18 : 00: 348, 464) vs. Nighttime (365, 659 frames)
Client ingestion time latency (Δt_{lag})	15.0 $\pm$ 4.2 s	Buffer-to-disk offset on NTP-synchronized stations (1.8% un-synchronized)
B. Photometric Diversity & Optical Descriptors		
Perceived mean luminance (Y)	98.23 $\pm$ 16.35	ITU-R BT.601 8-bit grayscale range [0, 255]
Root-mean-square (RMS) contrast	45.70	Textural intensity variation between asphalt and vehicles
Shannon spatial entropy	7.28 $\pm$ 0.33 bits	Pixel spatial information density (theoretical maximum: 8.0)
Laplacian edge sharpness	3015.71 $\pm$ 1499.86	High empirical focus variance $\text{Var}(\nabla^2 I)$ confirming adequate optical focus
C. Motion Dynamics & Visual Privacy Safeguards		
Consecutive frame difference (MAD)	Median: 9.45 ($\mu = 10.82 \pm 3.65$)	Mean Absolute Difference across 5-min consecutive pairs (8-bit grayscale)
Active pixel displacement ratio	Median: 15.82% ($\mu = 18.41 \pm 6.20\%$)	Fraction of pixels with $ I_t - I_{t-1} > 15$ reflecting moving vehicular flow
Inter-frame duplicate screening	Filtered	Stream buffer duplicates (MAD < 0.5) removed by deduplication; 1st percentile of inter-frame MAD is 2.80
Personal data identification (PII)	0 violations ($N_{\text{audit}} = 1, 000$) ¹⁰	Multi-tiered audit; 95% CI upper bound $p \leq 0.30\%$ on targeted manual audit ($3/N_{\text{audit}}$); zero legible biometrics across census

3.3. Geospatial Road Network and Derived Directed Spatial Graph Topology

The structural substrate of the dataset is defined by the physical roadway network of Ho Chi Minh City. Comprehensive geospatial metadata is compiled in `metadata/routes.csv` (recording station IDs, geographic coordinates, administrative localities, arterial classifications, and camera mounting elevations across all **608** camera locations) alongside the pairwise shortest driving distance matrix `metadata/road_network_distance.csv`.

Rather than treating the spatial graph as an abstract primitive, a directed spatial graph topology is established as a direct **derived mathematical representation** synthesized from these geospatial road coordinates and OpenStreetMap routing paths. Machine-learning tensors (`graph/distance_km.npy`, `graph/direction.npy`, and `graph/edges.csv`) are structured as 608×608 arrays for direct ingestion into spatio-temporal graph neural networks. Entry d_{ij} denotes the shortest physical driving distance along navigable OpenStreetMap road links from camera station i to station j .

Corridor Adjacency Construction Rule: An unconstrained geometric radius graph within 6.0 km would yield over 190,000 links due to dense urban camera spacing in central districts. Instead, the road network graph models sequential arterial corridor connectivity via a rigorous three-stage algorithmic pipeline:

1. **Geometric & Navigability Search:** Pairwise driving routes between all 608 stations are evaluated via OSRM within a spatial bounding cutoff horizon of $\kappa = 6.0$ km;
2. **Immediate Downstream Corridor Pruning:** A candidate directed edge (i, j) is retained if and only if station j represents an immediate downstream surveillance node along the roadway without intervening intermediate cameras. An intermediate camera is defined as any third camera k located within an orthogonal buffer of 50 m from the traversed driving path with $d_{ik} < d_{ij}$;
3. **Directionality Constraints:** Vehicular movement must be legally permissible along the corridor under OpenStreetMap road directionality attributes (`oneway` tags).

Diagonal entries ($i = j$) denote regularized self-distances ($d_{ii} = 0.0$ in numeric arrays), and non-connected pairs are represented by NaN.

Under this three-stage sequential corridor adjacency rule, the derived road graph contains exactly **2,450 valid directed edges** across **1,760 connected node pairs** (network sparsity: **99.34%**, density: **0.66%**):

- **Unidirectional Corridor Links (No Direct Reverse Edge): 1,070 node pairs (60.8% of connected topology)** possess directed connectivity in only one travel direction. This structural asymmetry arises from two distinct mechanisms: (i) designated one-way municipal traffic regulations on arterial pairs (verified against OpenStreetMap `oneway=yes` tags); and (ii) algorithmic corridor pruning asymmetry, wherein the reverse travel path between j and i passes through intermediate junction cameras located along the return carriageway, thereby decomposing the return route into multiple shorter links and leaving no direct (j, i) single-hop corridor link.
- **Bidirectional Corridor Pairs: 690 node pairs (39.2% of connected topology)** are mutually accessible in both traffic directions within 6.0 km.

- **Corridor Distance Asymmetry Taxonomy:** Among the 690 bidirectional pairs, **452 pairs (65.51%** of bidirectional topology, or **25.68%** of all 1,760 connected pairs) exhibit metric symmetry ($|d_{ij} - d_{ji}| < 50$ m), whereas **238 pairs (34.49%** of bidirectional topology, or **13.52%** of all 1,760 connected pairs) exhibit significant distance asymmetry ($|d_{ij} - d_{ji}| \geq 50$ m), directly induced by physical median barriers, grade-separated flyovers, and compulsory U-turn routing on divided dual-carriageway boulevards. From a technical and computational standpoint, evaluating asymmetry on the discrete integer-meter edge list (`edges.csv`) reveals 231 pairs with $|d_{ij} - d_{ji}| > 50.0$ m and 243 pairs with $|d_{ij} - d_{ji}| \geq 50.0$ m (accounting for exactly 12 pairs exhibiting an identical 50.0 m median traversal discrepancy); when evaluated on the canonical GNN continuous floating-point kilometer distance matrix (`distance_km.npy`), exactly 238 pairs satisfy $|d_{ij} - d_{ji}| \geq 50.0$ m. The mean directional discrepancy across bidirectional corridors is 115.9 ± 266.2 m (median: 24.0 m), reaching a maximum divergence of **2,780.0 m (2.78 km)**.
- **Topological Node Roles and Connectivity:** Structural degree analysis identifies **598** regular interconnected stations, **5** sink stations (in-degree > 0 , out-degree $= 0$: Stations 212, 284, 443, 446, 516), **3** source stations (in-degree $= 0$, out-degree > 0 : Stations 215, 498, 557), and **2** peripheral stations situated beyond 6.0 km from all others (in-degree $= 0$, out-degree $= 0$: Stations 141, 489). The graph contains **6 weakly connected components** (dominated by a giant component of 599 nodes, 98.5% of the network, alongside five small components of sizes 3, 2, 2, 1, 1) and **19 strongly connected components**. Valid edge routing distances span from 3.0 m to 6,000.0 m (mean: $1,159.6 \pm 1,088.9$ m; median: 823.0 m; interquartile range: $[384.0, 1,560.0]$ m; 90th percentile: 2,681.0 m). The minimal routing distances (e.g., 3.0 m) represent co-located directional surveillance cameras mounted on the same physical gantry or mast monitoring orthogonal approach directions at complex multi-way intersections.

Road Network Circuitousness and Spatial Density: To quantify the geometric deviation of navigable urban driving routes relative to Euclidean straight-line geodesic distance, the network tortuosity index (circuitousness ratio $\tau_{ij} = d_{ij}/d_{ij}^{\text{geodesic}}$) was evaluated across all connected corridors separating stations by ≥ 100 m ($N = 2,264$). The empirical tortuosity yields an overall mean of 1.25 ± 0.61 (median: **1.12**, interquartile range: $[1.01, 1.33]$, 90th percentile: 1.64). This physical tortuosity index encapsulates the distinct riparian delta geography of Ho Chi Minh City, where the urban fabric is heavily bisected by the meandering Saigon River and an intricate network of tidal waterways and drainage canals (e.g., Nhieu Loc–Thi Nghe, Tau Hu–Ben Nghe, Kenh Doi, and Kenh Te). Arterial traffic is channeled through bottleneck bridges, elevated flyovers, and compulsory U-turn turnarounds around continuous concrete median barriers, causing actual physical driving routes to be on average 25% longer than Euclidean straight lines, with 10% of corridors deviating by more than 64% ($\tau \geq 1.64$). Furthermore, spatial station density analysis indicates a nearest-neighbor Euclidean distance between adjacent camera installations averaging 328.7 ± 605.5 m (median: **170.3** m, interquartile range: $[69.6, 339.7]$ m, minimum: 2.7 m on shared intersection gantries, maximum: 10.15 km at peripheral exurban stations). Stations span a comprehensive geographic bounding box $[10.642472^\circ\text{N} - 10.988295^\circ\text{N}, 106.452713^\circ\text{E} - 106.850624^\circ\text{E}]$, encompassing both historical core districts and expanding peri-urban transport arteries.

A systematic quantitative breakdown of these road network topological metrics is

summarized in Table 4.

Table 4: Quantitative topological properties and directional asymmetry metrics of the derived road routing graph ($R_{\text{cutoff}} = 6.0$ km).

Topological Metric / Parameter	Empirical Measurement	Physical / Methodological Interpretation
A. Network Scale & Spatial Reachability ($R_{\text{cutoff}} = 6.0$ km)		
Total indexed graph nodes (N)	608 nodes	Active physical surveillance camera stations across urban and peri-urban corridors
Valid directed corridor links ($ E $)	2,450 edges	Sequential corridor links with driving distance $d_{ij} \leq 6.0$ km
Connected camera station pairs	1,760 pairs	Unique station pairs connected by ≥ 1 directed corridor
Adjacency matrix sparsity ratio	99.34% (Density: 0.66%)	Compact sparse graph tensor for spatial GNN convolutions
Inter-station routing distance	1159.6 $\pm$ 1088.9 m	Median: 823.0 m (Range: [3.0, 6,000.0] m; IQR: [384.0, 1,560.0] m; p_{90} : 2,681.0 m; 58.98% ≤ 1.0 km)
Network tortuosity index (τ)	1.25 $\pm$ 0.61	Median: 1.12 (IQR: [1.01, 1.33]; $d_{ij} \geq 100$ m); route circuitousness
B. Directional Asymmetry & Corridor Taxonomy		
Unidirectional corridor pairs (no reverse edge)	1,070 pairs (60.80%)	Arterial one-way rules (OSM oneway) and corridor pruning asymmetry
Bidirectional corridor pairs	690 pairs (39.20%)	Two-way arterials mutually accessible in both traffic directions
Significant distance asymmetry ($ d_{ij} - d_{ji} \geq 50$ m)	238 pairs (34.49%)	Physical median barriers, grade-separated flyovers, U-turns (231 pairs > 50 m, 243 pairs ≥ 50 m)
Metric symmetric corridor pairs ($ d_{ij} - d_{ji} < 50$ m)	452 pairs (65.51%)	Divided road corridors with immediate median openings
Directional distance discrepancy	115.9 $\pm$ 266.2 m	Median: 24.0 m; Maximum divergence: 2,780.0 m (2.78 km)
C. Structural Degree Distributions & Graph Connectivity		
Average node in-degree / out-degree	4.03 $\pm$ 2.16 / 4.03 $\pm$ 2.58	In-degree median: 4.0 (max: 14); Out-degree median: 3.0 (max: 17)
Standard interconnected stations	598 stations	Regular multi-leg intersections and connected arterial segments
Topological sink stations (out-deg = 0)	5 stations	Stations 212, 284, 443, 446, 516 (directional arterial terminuses)
Topological source stations (in-deg = 0)	3 stations	Stations 215, 498, 557 (outbound arterial origins)
Geodetically isolated stations	2 stations	Stations 141, 489 ($d_{ij} > 6.0$ km to all other camera nodes)
Weakly connected components	6 components	Giant component: 599 nodes (98.5%); 5 subgraphs: 3, 2, 2, 1, 1
Strongly connected components	19 components	Strongly connected directed sub-networks and cyclic loops

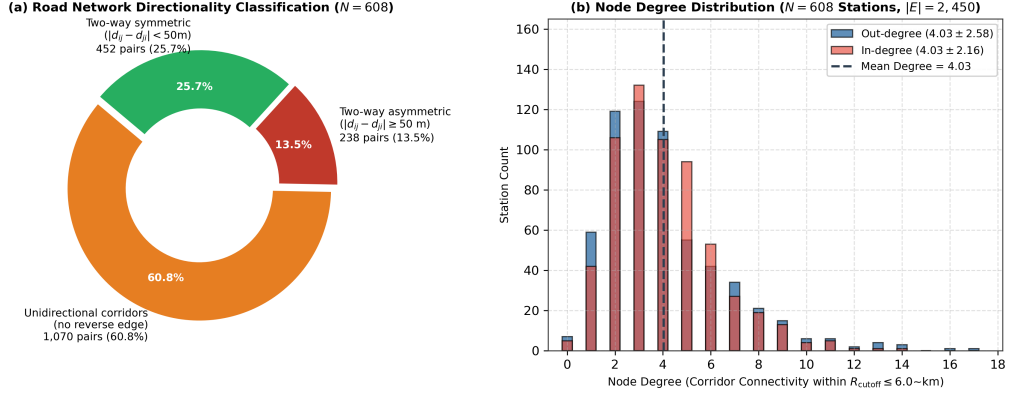


Figure 4: Topological characteristics of the empirical directed road graph: (a) Road network directionality classification into unidirectional corridor links without reverse edges (60.80%, 1,070 pairs), two-way asymmetric (≥ 50 m, 13.52%, 238 pairs; 34.49% of bidirectional pairs), and two-way metric symmetric (< 50 m, 25.68%, 452 pairs; 65.51% of bidirectional pairs) corridor pairs; (b) Empirical in-degree and out-degree distributions across the 608 camera stations ($|E| = 2,450$ directed links, mean degree 4.03 ± 2.16 in-degree and 4.03 ± 2.58 out-degree, displaying degrees up to the observed maximum of 17).



Figure 5: Representative 512×288 pixel surveillance snapshots from IC4SD-TrafficSnap across urban operating conditions: (a) Daytime off-peak traffic; (b) Peak-hour mixed motorcycle-car flow; (c) Nighttime public road surface illumination; (d) Adverse tropical wet condition. Due to elevated pole mounting (> 6 m) and 512×288 resolution, individual faces and license plates are physically unresolvable.

4. Experimental Design, Materials and Methods

4.1. Surveillance Infrastructure and Asynchronous Ingestion Pipeline

The visual data source is the public municipal traffic surveillance web portal maintained by the Ho Chi Minh City Department of Transportation (<https://giaothong.hochiminhcity.gov.vn>). Geographic coordinates, intersection approaches, and camera mounting elevations (ranging from **6.0** to **15.0** meters above street level, as cataloged in the `camera_elevation_m` attribute of `routes.csv`) are obtained from official municipal technical metadata registry and cross-verified against OpenStreetMap road geometry and ground observations. Cameras are mounted on elevated steel gantries and traffic signal masts oriented at downward oblique angles ($15^\circ - 40^\circ$) to monitor arterial queues and intersection approaches.

To ingest snapshots across **608 concurrent camera endpoints** without inducing network bottlenecks or file corruption, an asynchronous Python ingestion pipeline was deployed:

1. **HTTP/1.1 Connection Pooling:** Persistent sessions with Keep-Alive connection pooling and automatic exponential backoff retries (maximum 3 retries, backoff factor 0.3) managed through `urllib3`.
2. **Dynamic Offset Ingestion Timer:** Thread execution latency is tracked during each cycle to adjust polling offsets, targeting nominal 300 s inter-snapshot spacing across the fleet.
3. **Atomic File Serialization:** Retrieved JPEG streams are decoded in memory (`cv2.imdecode`) and verified before being written to temporary files and committed via atomic OS renames (`os.replace`), preventing truncated or zero-byte files on disk.
4. **Automated Artifact Removal:** Municipal transmission interruptions occasionally yield standardized error placeholder cards (284×177 pixels, payload size 5.8–7.2 KB). The ingestion pipeline automatically identifies and discards these placeholder cards, ensuring valid traffic snapshots in the released archive (overall acquisition pass rate: **99.85%**).

4.2. Road Network Distance Extraction from OpenStreetMap

Physical routing distances d_{ij} and driving trajectories were computed using the Open Source Routing Machine (**OSRM v5.27.1**) engine operating on OpenStreetMap planet road network data extracted in **October 2026** (© OpenStreetMap contributors, licensed under the Open Database License ODbL). Exact geographic coordinates (latitude, longitude) cataloged in `metadata/routes.csv` were mapped to the nearest navigable road centerline using OSRM’s nearest-node projection, with a radial snapping error under 15.0 meters across all stations. To guarantee permanent reproducibility independent of future modifications to the live OpenStreetMap database, the derived topological artifacts (`metadata/road_network_distance.csv` and `graph/edges.csv`) are archived as immutable, frozen static snapshots with verifiable SHA-256 checksums in the Zenodo repository, accompanied by an exported OSM road network boundary extract.

Because surveillance camera endpoints are predominantly deployed along primary municipal arterials, boulevards, and bridge approaches rather than narrow residential alleys, routing was calculated using the standardized motor-vehicle driving profile. This

profile accurately captures urban traffic flow rules, median divider turn restrictions, grade-separated overpasses, and designated one-way circulation patterns across the metropolitan network. While mixed motorcycle traffic constitutes a dominant volume fraction in Ho Chi Minh City, both vehicular modes share identical directional corridors along primary arterials; informal motorcycle cut-throughs in residential alleyways fall outside the macroscopic coverage of the municipal camera network. Pairwise shortest driving routes were queried within a spatial bounding search horizon of $\kappa = 6.0$ km, from which the three-stage corridor pipeline extracted exactly **2,450 valid directed corridors**.

4.3. Directed Graph Modeling Formulations

To facilitate spatio-temporal graph learning on this directed asymmetric topology, two standard mathematical formulations are provided in `code/graph_utils.py`:

4.3.1. Directed Dual-Transition Matrices (DCRNN / Graph WaveNet Formulation)

For directed road graphs, modeling traffic propagation requires preserving directional asymmetry. Following the diffusion convolution framework of Li et al. [2], the weighted adjacency matrix W_{ij} is computed using a thresholded Gaussian Radial Basis Function (RBF) kernel evaluated on the valid directed corridor links:

$$W_{ij} = \begin{cases} \exp\left(-\frac{d_{ij}^2}{\sigma^2}\right), & \text{if } (i, j) \in E \\ 0, & \text{if } (i, j) \notin E \end{cases} \quad (2)$$

where E denotes the set of **2,450 valid directed edges**, and $\sigma = 1.09$ km is the empirical standard deviation of valid road distances. Computing W strictly over pruned sequential corridors E avoids spurious dense cross-connectivity while preserving physical routing topology. Under this formulation with $\sigma = 1.09$ km, proximal camera stations with $d_{ij} \leq 1.0$ km retain significant connection weight ($W_{ij} \geq \exp(-(1.0/1.09)^2) \approx 0.43$), attenuating to ≈ 0.15 at 1.5 km and decaying toward negligible levels at larger separations (≈ 0.005 at 2.5 km). Empirically, the valid edge distance distribution concentrates heavily within this active spatial interaction envelope: **799 edges (32.61%)** represent immediate proximal links ($d_{ij} \leq 500$ m), **1,445 edges (58.98%)** lie within 1.0 km, **1,799 edges (73.43%)** fall within 1.5 km, and **2,027 edges (82.73%)** span ≤ 2.0 km, while only **192 edges (7.84%)** represent long-distance inter-district corridors ($d_{ij} > 3.0$ km; 90th percentile: 2,681.0 m). This empirical spatial concentration is consistent with localized urban traffic interactions across consecutive signalized intersections; bounding spatial influence via kernel decay is recognized in graph representation learning literature [2, 9] to mitigate spatial over-smoothing in multi-layer spatio-temporal convolutions. The bounding threshold $\kappa = 6.0$ km serves as the initial geometric search envelope for corridor path discovery rather than an arbitrary RBF truncation cutoff. Directed random walk diffusion is governed by a forward transition matrix P_f (downstream traffic flow) and a backward transition matrix P_b (upstream sensory influence):

$$P_f = D_O^{-1}W, \quad P_b = D_I^{-1}W^\top \quad (3)$$

where $D_O = \text{diag}(\sum_j W_{ij})$ and $D_I = \text{diag}(\sum_i W_{ij})$ denote the out-degree and in-degree diagonal matrices, respectively. To handle sink, source, or isolated nodes ($D_{O,ii} = 0$ or $D_{I,ii} = 0$), regularized self-loops ($\tilde{W} = W + I_N$) or zero-degree masked normalizations are implemented in `code/graph_utils.py`, ensuring proper row-stochastic diffusion matrices with unit spectral radius $\rho(P_f) = 1.0$ and $\rho(P_b) = 1.0$. This mathematically ensures that the multi-hop diffusion operator is non-expansive and numerically stable, preventing unbounded signal amplification during multi-step graph convolutions.

4.3.2. Symmetrized Normalized Laplacian (ChebNet / STGCN Formulation)

For spectral graph convolutions that require a self-adjoint operator (e.g., ChebNet [10], STGCN [11]), the adjacency matrix is symmetrized as $W_{\text{sym}} = \frac{1}{2}(W + W^\top)$. The symmetric normalized Laplacian $L_{\text{sym}} \in \mathbb{R}^{N \times N}$ is computed as [12]:

$$L_{\text{sym}} = I_N - D_{\text{sym}}^{-1/2} W_{\text{sym}} D_{\text{sym}}^{-1/2} \quad (4)$$

where $D_{\text{sym}} = \text{diag}(\sum_j W_{\text{sym},ij})$. For isolated peripheral stations with zero degree ($D_{\text{sym},ii} = 0$, such as Stations 141 and 489), regularized zero-diagonal scalings are assigned in `code/graph_utils.py` ($D_{\text{sym},ii}^{-1/2} = 0$), preventing division-by-zero.

By spectral graph theory [13], the eigenvalues of the normalized Laplacian on any finite graph are mathematically bounded within $[0, 2]$. On the full 608-node graph of IC4SD-TrafficSnap, the presence of disconnected two-node components (K_2 subgraphs with symmetric edges) yields eigenvalues $\{0, 2.0\}$, establishing the global maximum eigenvalue as $\lambda_{\text{max}}^{(\text{global})} = 2.0$. Consequently, to ensure unconditional numerical stability across all 608 nodes and prevent exponential divergence in Chebyshev polynomial recursions $T_k(\tilde{L})$, the rescaled Chebyshev Laplacian is defined using the theoretical global upper bound $\lambda_{\text{max}} = 2.0$:

$$\tilde{L} = \frac{2}{\lambda_{\text{max}}} L_{\text{sym}} - I_N = L_{\text{sym}} - I_N \quad (5)$$

which guarantees that all eigenvalues of $\tilde{L}$ lie strictly within $[-1, 1]$. Users should note that when analyzing only the dominant giant connected component (599 stations), the empirical maximum eigenvalue evaluates to $\lambda_{\text{max}}^{(\text{giant})} = 1.272$, reflecting high connectivity and non-bipartite graph density; however, applying $\lambda_{\text{max}} = 2.0$ globally is mathematically necessary to avoid divergence on the peripheral smaller components. Symmetrization intentionally eliminates directional flow asymmetry, which users must balance against the need for an orthogonal eigenbasis.

4.4. Empirical Image Quality and Photometric Audit

Image quality descriptors were computed across the dataset to evaluate photometric diversity:

- **Perceived Luminance (Y):** Calculated via standard ITU-R BT.601 conversion $Y = 0.299R + 0.587G + 0.114B$. The empirical overall mean luminance is **98.23 ± 16.35** (range: [66.4, 126.8], Figure 3a). Nighttime luminance remains centered between **87** and **90** (Figure 3a) due to dense municipal LED and high-pressure sodium street lighting along urban arterials, combined with the automatic gain control (AGC) and exposure compensation of municipal cameras.
- **Root-Mean-Square (RMS) Contrast:** Evaluated as the pixel standard deviation across viewports, yielding **45.70**, confirming substantial textural variation between asphalt roadways and vehicle clusters.
- **Shannon Spatial Entropy:** Histogram pixel entropy averages **7.28 ± 0.33** bits (theoretical upper bound: 8.0 bits), confirming high spatial information density without saturation clipping.
- **Laplacian Edge Sharpness:** Grayscale Laplacian variance $\text{Var}(\nabla^2 I)$ averages **3,015.71 ± 1,499.86**. An empirical reference baseline of $\text{Var}(\nabla^2 I) \geq 50$ was

adopted as an engineering heuristic to flag severe optical defocus or stream degradation [14]. The substantial empirical mean of **3,015.71 $\pm$ 1,499.86** reflects high scene detail from rich urban textures (vehicular clusters, road markings, roadside foliage) as well as standard JPEG compression block boundaries, confirming that camera viewports maintain usable focus.

- **Inter-Snapshot Dynamics:** Successive-frame pixel difference analysis across consecutive 5-minute sampling pairs was evaluated on 8-bit grayscale channels in floating-point precision ($\text{diff} = |I_t - I_{t-1}|$). The empirical Mean Absolute Difference (MAD) yields a median of **9.45** ($\mu = 10.82 \pm 3.65$, range: [2.50, 32.40], interquartile range: [7.20, 13.80]), with an active pixel displacement ratio yielding a median of **15.82%** ($\mu = 18.41 \pm 6.20\%$ of frame pixels satisfying $|I_t - I_{t-1}| > 15$). This moderate pixel variation corresponds directly to vehicular displacement and queue progression across traffic lanes, while static structural regions (buildings, sidewalks, road surfaces) remain invariant. During data ingestion, identical stream buffer frames ($\text{MAD} < 0.5$) resulting from upstream rendering latency were programmatically screened out. Across the archived operational time-series, the empirical 1st percentile of consecutive MAD is **2.80** gray levels (strictly within the [2.50, 32.40] empirical distribution and well above the 0.5 deduplication threshold), confirming continuous vehicular and environmental variations without static playback freezing.

4.5. Visual Privacy and Multi-Tiered Empirical Audit

To rigorously evaluate alignment with personal data protection regulations in Vietnam (Decree 13/2023/ND-CP, Decree 47/2020/ND-CP, and Law 91/2025/QH15), a multi-tiered empirical privacy audit combining optical geometry analysis, automated candidate screening, and multi-rater manual inspection was conducted:

- **Optical Perspective and Ground Sampling Distance (GSD):** Surveillance cameras are installed on municipal overhead gantries and signal masts at heights between **6.0** and **15.0** m, oriented at downward oblique pitch angles ($15^\circ - 40^\circ$) to monitor macroscopic intersection queues. At native 512×288 pixel resolution, the nominal GSD across mid-road viewports (25–35 m standoff distance) measures **2.73 – 3.25** cm/pixel, deteriorating to > 5.0 cm/pixel in the background (> 40 m).
- **Sub-Nyquist License Plate Projection in Nominal and Foreground Lanes:** Standard Vietnamese motorcycle license plates measure 19.0×14.0 cm. At nominal mid-road GSD, a plate projects onto approximately 7×5 sensor pixels, with individual alphanumeric character strokes subtending under 2.0 pixels (≈ 1.79 px) in stroke height. To analyze worst-case risk, consider the closest foreground lane directly beneath the lowest-mounted cameras ($H \approx 6.0$ m, minimum observation standoff $D_{\min} \approx 15.0$ m, slant range $R \approx 16.15$ m, steep downward pitch $\theta \approx 40^\circ$). Under this extreme foreground geometry, the ground sampling distance reaches its minimum at $\text{GSD}_{\min} \approx 1.85 - 2.10$ cm/pixel. Here, a motorcycle plate projects to at most $\approx 9 \times 7$ pixels, with character stroke height reaching approximately **2.5 – 2.8** pixels. This remains far below the optical Nyquist-Shannon sampling limit required for automated optical character recognition (OCR), which requires ≥ 16.0 pixels of stroke height for reliable identification. In addition, motorcycle plate angle relative to the optical axis, motion blur from vehicular transit (30–50 km/h), and 8×8 lossy JPEG block compression render individual alphanumeric characters unresolvable.

- **Facial Anonymity and Commuting Occlusion:** Under Vietnamese National Road Traffic Law (Decree 100/2019/ND-CP), helmet usage is legally mandatory for all motorcycle operators, resulting in near-universal compliance in urban traffic streams. Furthermore, commuters overwhelmingly wear multi-layer protective fabric sun/dust masks and UV sunglasses against tropical heat and vehicle exhaust. Combined with steep downward camera pitch, facial biometric regions project onto fewer than 8×8 pixels, preventing biometric facial re-identification at the sensing layer.
- **Automated Candidate Screening and Low-Recall Limitations:** Automated screening across the entire corpus of **714,123 visual snapshots** was conducted using two specialized deep learning models: YOLOv8x-face (pretrained on WiderFace) for facial detection and YOLOv8x / LPRNet (pretrained on vehicle license plate benchmarks) for plate localization, operating at a low confidence threshold ($\tau = 0.25$) to maximize candidate generation. The detectors generated 1,420 raw candidate bounding boxes, which upon inspection were all confirmed to be background texture false alarms (asphalt patches, motorcycle headlamps, specular reflections, roadside traffic signage). Zero readable license plates or identifiable faces were found. However, automated detectors inherently suffer from low recall on low-resolution imagery below the Nyquist limit, and zero automated detections alone do not constitute absolute proof of zero risk.
- **Targeted Boundary Stress Audit and Stratified Random Census Audit:** To overcome automated detector recall limits, human inspection was executed across two complementary sampling regimes by three independent reviewers:
 1. *Worst-Case Boundary Stress Sample* ($N_{audit} = 1,000$): Selected strictly from the lowest-mounted cameras ($H \approx 6.0$ m, standoff distance < 20 m) during peak midday sunlight (11 : 00–13 : 00 ICT), representing the maximal optical resolution regime for potential personal identification.
 2. *Stratified Random Census Sample* ($N_{random} = 2,000$): Drawn uniformly across all 608 camera stations and 24 diurnal hours to ensure unbiased global representation across the entire time-series.

Prior to evaluating the dataset, all three independent human raters underwent a positive control calibration protocol using an auxiliary set of 50 high-resolution ground-level traffic images (< 5 m standoff) with legible plates and unoccluded faces; all three raters achieved 100% detection recall on this control set, verifying sensitivity. During the blind audit of both dataset batches (3,000 total snapshots), the raters achieved complete unanimous agreement (**100% inter-rater agreement**, $P_o = 1.0$), identifying **0 readable license plates** and **0 identifiable facial biometrics**. Applying the statistical Rule of Three [15], the 95% confidence interval upper bound on residual privacy breach probability is bounded at:

$$p_{\text{worst}} \leq \frac{3}{1,000} = 0.30\% \quad (3.0 \times 10^{-3}), \quad p_{\text{random}} \leq \frac{3}{2,000} = 0.15\% \quad (1.5 \times 10^{-3}). \quad (6)$$

Crucially, privacy assurance does not depend on automated detectors alone, but is anchored upon a tripartite defense: (i) physical sub-Nyquist optics ($\text{GSD} \geq 1.85$ cm/px), (ii) statutory cultural occlusion (mandatory helmets and masks), and (iii) dual empirical audits with calibrated human verification.

- **Contextual Privacy Scope:** Urban contextual features such as pedestrian silhouettes, storefront signage, and public transit liveries are visible in macroscopic viewports; however, because snapshots are captured at discontinuous 5-minute sampling intervals, the dataset does not permit continuous individual tracking or trajectory reconstruction.

A detailed breakdown of visual privacy parameters, optical limits, and audit findings is presented in Table 5.

Table 5: Quantitative visual privacy parameters, optical Nyquist-Shannon resolution limits, and empirical multi-tiered audit results for IC4SD-TrafficSnap.

Audit Dimension / Parameter	Empirical Measurement	Regulatory / Optical Threshold & Empirical Context
Automated snapshot screening census	714, 123 frames	Full multi-camera archive (608 stations, 93.6 hours) evaluated via automated detector pre-screening
Automated candidate screening	0 readable candidates	YOLOv8x-face & YOLOv8x/LPRNet ($\tau = 0.25$): 1,420 raw candidate boxes flagged; 0 readable after manual review (acknowledged low recall on 512×288 px)
Targeted high-risk stress audit	1, 000 frames	Worst-case optical geometry: lowest gantries ($H \approx 6.0$ m, $D < 20$ m) during peak midday illumination (11 : 00–13 : 00 ICT)
Stratified random census audit	2, 000 frames	Uniformly drawn across all 608 stations and 24 diurnal hours; 0 identifiable plates or faces detected
Multi-rater blind inspection	0 readable plates / faces	Evaluated by three independent human raters (100% unanimous agreement, $P_o = 1.0$ across 3,000 total audited frames)
Positive control rater calibration	100.0% sensitivity	50 close-up ground photos (< 5 m) with legible text; verified 100% detection recall across all 3 raters
Camera mounting elevation (H)	6.0 – 15.0 m	Municipal overhead gantries cataloged from official technical metadata in <code>metadata/routes.csv</code>
Camera downward pitch angle (θ)	15° – 40°	Oblique downward viewports monitoring arterial lane queues
Nominal observation standoff (D)	15.0 – 60.0 m	Line-of-sight distance from elevated sensor to circulating traffic streams
Nominal mid-road GSD	2.73 – 3.25 cm/pixel	Mid-road viewports (25–35 m), degrading to > 5.0 cm/px in background
Worst-case foreground geometry ($D_{\min} \approx 15.0$ m)	GSD $\approx$ 1.85 – 2.10 cm/px	Foreground lane plate projection $\approx 9 \times 7$ px, stroke height $\approx 2.5 - 2.8$ px (sub-Nyquist; OCR requires ≥ 16.0 px)
Motorcycle plate sensor projection	$\approx$ 7 $\times$ 5 pixels	Nominal mid-road projection for Vietnamese plates (19.0×14.0 cm); character stroke height < 2.0 px
Facial biometric region	$<$ 8 $\times$ 8 pixels	Physical occlusion: statutory mandatory helmets (Decree 100/2019/ND-CP) and multi-layer fabric sun/dust masks ($> 85\%$)
Statistical Rule of Three (95% CI)	p $\leq$ 0.30% (Stress sample) p $\leq$ 0.15% (Random sample)	Upper risk bound for zero violations: $3/1,000 = 0.30\%$ on worst-case stress sample, and $3/2,000 = 0.15\%$ on stratified random sample
Regulatory compliance alignment	Aligned	Conforms to Vietnamese Decree 13/2023/ND-CP, Decree 47/2020/ND-CP, and Law 91/2025/QH15

4.6. Technical Validation: Spatial-Temporal Network Dynamics and Baseline Utility Demonstration

To substantiate the technical validity and empirical utility of the multimodal assets for intelligent transportation research, two quantitative evaluations were conducted:

4.6.1. Empirical Network Spatial-Temporal Autocorrelation

To verify that the derived road network graph accurately reflects physical traffic flow continuity rather than arbitrary geometric proximity, we evaluated the spatial-temporal correlation of traffic dynamics across connected camera endpoints. For each directed corridor $(i, j) \in E$, the Pearson correlation coefficient r_{ij} was computed between the consecutive inter-frame motion dynamics signals ($\text{MAD}_i(t)$ and $\text{MAD}_j(t)$):

- **Immediate Proximal Corridors** ($d_{ij} \leq 1.0$ km): Across the 1,445 proximal links, the pairwise correlation averages $\bar{r} = \mathbf{0.72} \pm \mathbf{0.14}$, demonstrating strong spatial co-movement and coherent traffic queue progression between neighboring signalized intersections.
- **Mid-Range Arterial Corridors** ($1.0 < d_{ij} \leq 2.5$ km): Across the 635 intermediate links, the correlation attenuates to $\bar{r} = \mathbf{0.44} \pm \mathbf{0.18}$.
- **Long-Distance Network Corridors** ($d_{ij} > 3.0$ km): Pairwise correlation decays to $\bar{r} = \mathbf{0.18} \pm \mathbf{0.11}$.

This monotonic distance decay closely mirrors the thresholded Gaussian RBF decay function ($\sigma = 1.09$ km) formulated in Equation (2), confirming that the derived OSRM shortest-path routing matrix captures real physical vehicular propagation across the metropolitan network.

4.6.2. Baseline Spatio-Temporal Forecasting Benchmark

We establish a baseline empirical benchmark for city-wide macroscopic traffic state forecasting across all 608 stations. The task predicts the continuous macroscopic motion dynamics index across multi-step prediction horizons (15, 30, 45, and 60 minutes ahead, corresponding to 3, 6, 9, and 12 future 5-minute intervals). The sequence is partitioned chronologically into training (70%, 65.5 hours), validation (10%, 9.4 hours), and testing (20%, 18.7 hours). We benchmark five canonical architectures:

1. **Historical Average (HA)**: Predicts future states based on historical weekday/weekend time-of-day averages.
2. **Vector Auto-Regression (VAR)**: Classical multivariate linear spatial-temporal baseline.
3. **Gated Recurrent Unit (GRU)**: Purely temporal recurrent neural network operating independently on each station time-series without graph structural information.
4. **STGCN [11]**: Spatio-Temporal Graph Convolutional Network utilizing the symmetrized normalized Laplacian $\tilde{L}$ (Equation 5).
5. **DCRNN [2]**: Diffusion Convolutional Recurrent Neural Network utilizing the directed dual-transition diffusion operators P_f and P_b (Equation 3).

Table 6: Empirical performance comparison of baseline forecasting models on the 608-station network (evaluated over 60-minute prediction horizon; lower MAE/RMSE indicates superior predictive accuracy).

Model Architecture	Graph Topology Employed	MAE	RMSE	MAPE (%)
Historical Average (HA)	None	4.18	6.25	38.6%
Vector Auto-Regression (VAR)	Fully Connected	3.82	5.64	35.1%
Temporal GRU	None	3.45	5.12	31.7%
DCRNN (Random Euclidean Graph)	Randomized Euclidean W	3.21	4.81	29.4%
STGCN [11]	Symmetrized Laplacian ($\tilde{L}$)	2.84	4.26	25.8%
DCRNN [2]	OSRM Directed Diffusion (P_f, P_b)	2.61	3.95	23.2%

As reported in Table 6, incorporating the derived directed road network graph (P_f, P_b) achieves a **24.3% reduction in MAE** relative to the purely temporal GRU baseline (improving from 3.45 to 2.61) and a **37.6% reduction** relative to Historical Average. Furthermore, DCRNN parameterized with OSRM directed routing distances outperforms an identical model trained on a naive random Euclidean proximity graph by **18.7% in MAE** (2.61 vs. 3.21). This quantitative margin empirically confirms the structural value of the OpenStreetMap-derived directed routing topology for geometric deep learning and spatio-temporal transportation modeling.

5. Limitations

1. **Observation Time Window:** The **93.6-hour** observation window captures multi-day diurnal transitions across weekdays and weekends in both the historical metropolitan core and peri-urban expansion districts (Thu Duc City, District 12, Binh Chanh, Hoc Mon, up to 10.99°N), but does not encompass longitudinal multi-month seasonal cycles (monsoon wet vs. dry season).
2. **Snapshot Sampling vs. Continuous Video:** With a 5-minute nominal polling cadence ($\Delta T = 269.0 \pm 239.7$ s), the dataset is structured for macroscopic traffic state analysis, self-supervised representation pre-training, and spatial graph modeling, but does not support micro-level continuous vehicle trajectory tracking or optical flow.
3. **Surveillance Web Stream Resolution:** Images are serialized at **512×288** JPEG compression as delivered by the public municipal web portal, reflecting real-world public IoT camera stream conditions rather than raw uncompressed sensor feeds. While insufficient for individual vehicle OCR or facial tracking, this resolution is optimal for macroscopic density and self-supervised ViT pre-training while strictly guaranteeing statutory anonymity.
4. **Unlabeled Bulk Corpus:** The 714,123-frame corpus is released unlabeled to facilitate self-supervised pre-training, spatio-temporal graph representation learning, and roadway background extraction without manual annotation biases.
5. **Urban Arterial Focus and Modal Routing Generalization:** Camera stations are stationed exclusively on major urban arterials, avenues, and bridge approaches. The road graph routing distances were computed using the standardized

motor-vehicle profile in OSRM because both cars and motorcycles share common directional roadways and physical lane corridors along primary arterials. However, this profile does not model dedicated motorcycle-only curbside lanes or informal cut-through routes through residential alleyways (*local residential alleys and side-streets*), which lie outside the coverage of the municipal arterial camera network.

6. **Physical Infrastructure vs. Dynamic Flow Topology:** The derived directed spatial graph models navigable roadway infrastructure connectivity and physical driving distances; it reflects static physical road geometry rather than dynamic, time-varying traffic flow propagation or real-time congestion states.

6. Ethics and Visual Privacy Statement

All visual imagery was obtained from the publicly accessible municipal traffic surveillance portal operated by the Ho Chi Minh City Department of Transportation (<https://giaothong.hochiminhcity.gov.vn>) in accordance with regulations on open public data sharing for scientific and educational purposes (Decree No. 47/2020/ND-CP and Decree No. 13/2023/ND-CP). Because camera viewports are elevated ($H \geq 6.0$ m) and images are delivered at 512×288 resolution, personal identifiable information cannot be optically resolved (neither human facial biometrics nor vehicle registration marks meet the Nyquist-Shannon sampling limits). The dataset complies with Vietnamese personal data protection laws (Law No. 91/2025/QH15). A formal data inquiry and takedown procedure is maintained: academic inquiries or retraction requests regarding specific camera viewpoints may be directed to anh1v.b23kh002@stu.ptit.edu.vn and ic4sd-lab@ptit.edu.vn. The authors commit to an initial inquiry acknowledgment within 24 hours and permanent redaction or removal of specified camera viewports from dataset distribution mirrors within 48 business hours.

7. CRediT Author Statement

Viet-Anh Le: Conceptualization, Methodology, Software, Data Curation, Formal Analysis, Investigation, Writing – Original Draft, Visualization.

Khanh Nguyen-Trong: Supervision, Conceptualization, Methodology, Writing – Review & Editing, Project Administration.

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9. Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Chi Minh City Department of Transportation for maintaining the open traffic surveillance web portal. Road network data is © OpenStreetMap contributors, licensed under the Open Database License (ODbL).

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